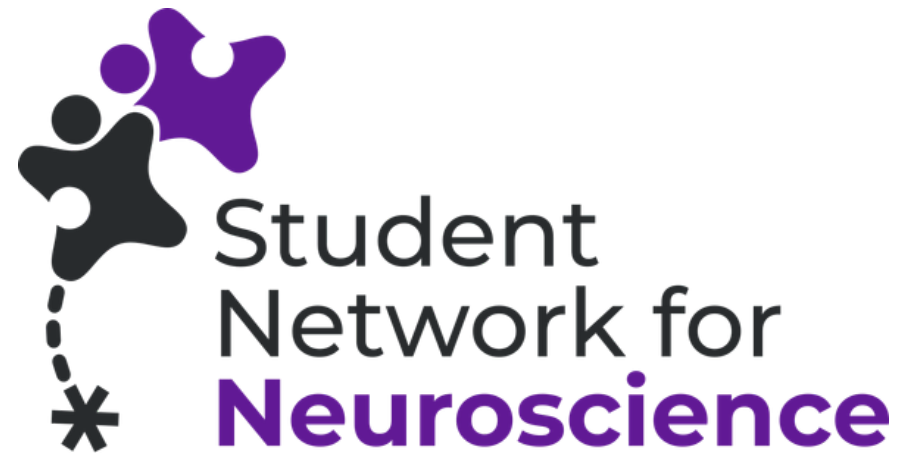
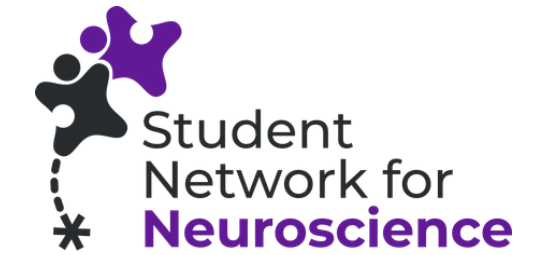




Behavior & Brain Waves



Contents



- Recap: Go/NoGo-Paradigm
- Behavioral Data Analysis
- ERPs
- Topographies
- *Statistical Analysis (as basis for interpretation)*
- Interpretation; Relation between Behavior and EEG

Recap



Load &
Merge

Filter

Re-ref
& ICA

Extract
Epochs

Baseline
& Reject

ERPs &
Topographies

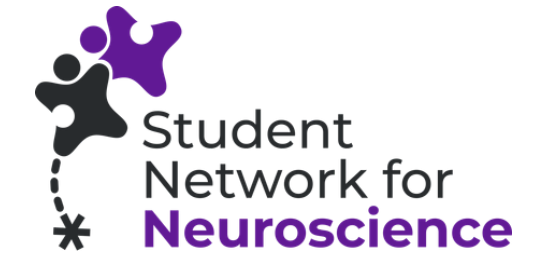
Paradigm: Go/No-Go Task

Participants pressed a button for 'Go' stimuli and withheld their response for 'No-Go' stimuli.

Epochs will be time-locked to stimulus onset

We will compare Go vs. NoGo to examine N2 and P3 amplitude differences across conditions

Paradigm



- simple experimental paradigm
- go condition: response by pressing button
- no-go condition: withhold automatic response

=> measure response inhibition

<https://www.testable.org/experiment-guides/executive-function/go-no-go-task>

Paradigm - example

- participants get instructions
- in every trial presentation (e.g. 1s) of either...
 - yellow square => press button in time as fast as possible (e.g. 1s to respond)
 - or...
 - blue square => no response
- most trials are go trials => create an automatic response behaviour
- occasional no-go trials => challenging response inhibition
- important: relative likelihood of go and no-go trials

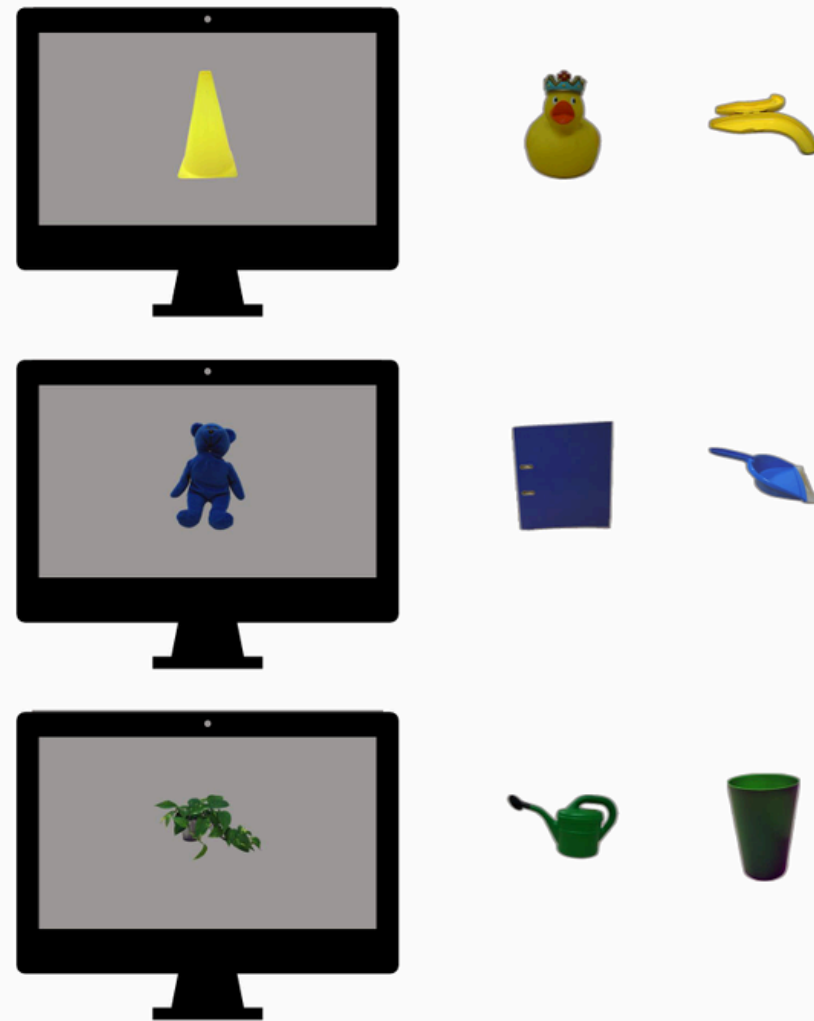
<https://www.testable.org/experiment-guides/executive-function/go-no-go-task>



https://www.testable.org/wp-content/uploads/2022/11/Go_nogo_illustration.png

Stimuli we use

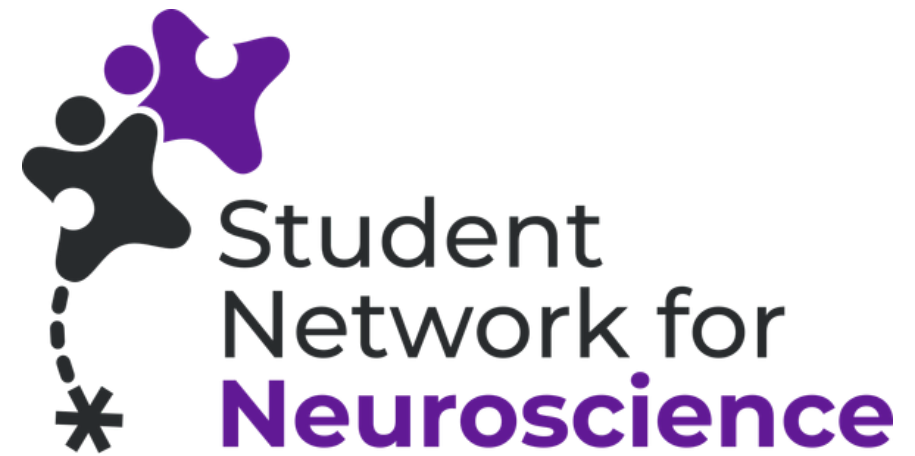
GO-Trials



NOGO-Trials



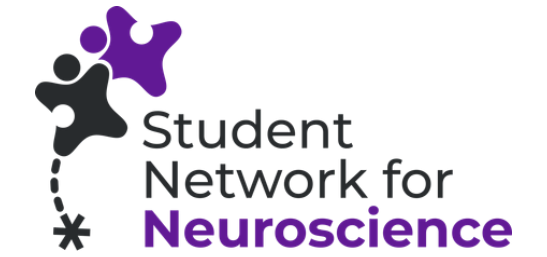
Seminar Programmieren 25 - Allgemeine Psychologie I



Behavior



Behavioral Data Analysis

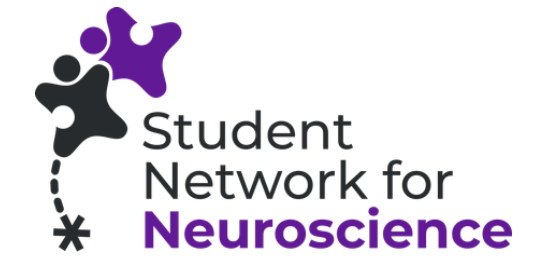


measures for behavioural analysis:

- key press and reaction times (RT) => analyse accuracy (hit rate), error rate, sensitivity, specificity, etc
- which one ? => depends on study/hypotheses
- context for interpretation

slides Merle Sagehorn (University of Osnabrück)

Behavioral Data Analysis



Key outcome measures

Commission Errors (CE)

Proportion of No-Go responses not successfully withheld = false hit/alarm.

High CE → poor response inhibition.

Omission Errors (OE)

Failure to respond to a Go stimulus = false miss.

Reflects lapses in attention / vigilance.

Mean Reaction Time (RT)

Encompasses vigilance, decision-making, response initiation & preparedness.

SD of Reaction Time

Variability in RT; higher SD = more inconsistent performance.

Sensitivity d'

Signal-detection: distinguishes signal from noise.

Key measure of inhibition and recognition abilities

→ Higher d' = better ability to discriminate.

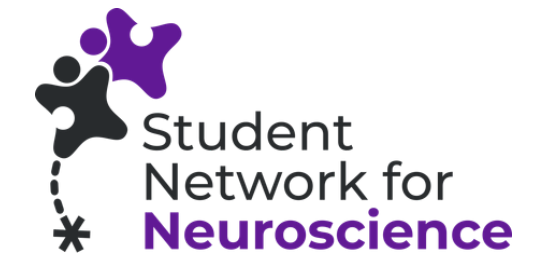
Response Criterion β

Reflects response bias: cautious vs. impulsive style.

Low β = liberal; High β = conservative.

Wright et al. (2014). Response inhibition and psychopathology: a meta-analysis of go/no-go task performance. Journal of Abnormal Psychology, 123(2), 429.

Behavioral Data Analysis



Analysis Pipeline

Load behavioral data - using *glob*-library to find the files

Combine the data to one dataframe - using *panda*-library and the `pd.concat` function

(Clean data from PsychoPy issues - delete contaminating lines)

Calculate Accuracy (Hits vs. correct rejections)

1. Group the dataset by condition (Go vs. NoGo)
2. Calculate the mean accuracy (percentage) for both conditions.
3. Plot the results in a simple bar chart.

Filter for & compare Reaction Times for Hits & False Alarms

```
df = df_raw.dropna(subset=['type']).copy()
df = df[df['type'].isin(['go', 'nogo'])].copy()

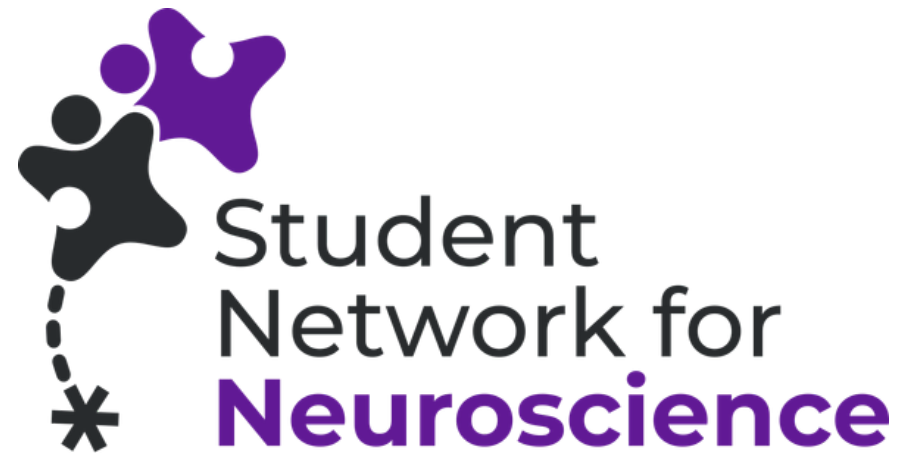
print(f"Left: {len(df)}")
print(f"Deleted: {len(df_raw) - len(df)} - lines that contaminate the data (Intro, Delay, Breaks, Outro)\n")

df[['type', 'corrAns', 'TrialResp.keys', 'TrialResp.corr', 'TrialResp.rt']].head(10)

# Look for Number of Go vs. NoGo Trials (should be around 75/25)
print(df['type'].value_counts())
print(f"\nTotal: {len(df)} Trials")

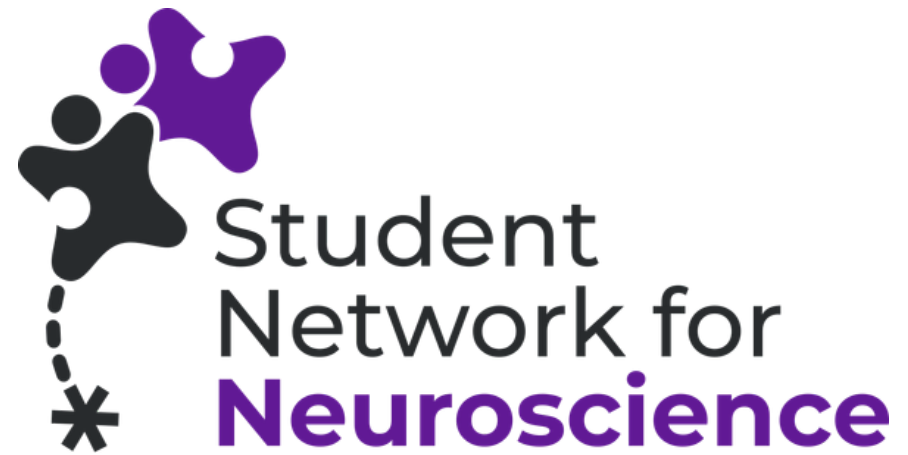
# Check: Wo stehen die RT-Daten wirklich?
print("TrialResp.rt vorhanden:", df['TrialResp.rt'].notna().sum())
print("TrialLoop.TrialResp.rt vorhanden:", df['TrialLoop.TrialResp.rt'].notna().sum())
print()
print("TrialResp.corr vorhanden:", df['TrialResp.corr'].notna().sum())
print("TrialLoop.TrialResp.corr vorhanden:", df['TrialLoop.TrialResp.corr'].notna().sum())
```

```
# Define Go and NoGo trials based on the 'corrAns' column
# If the correct answer was 'None' or empty, it was a NoGo trial!
df['Condition'] = df['corrAns'].apply(
    lambda x: 'NoGo' if pd.isna(x) or str(x).strip().lower() in ['none', ''] else 'Go'
)
```



Any questions so far?





ERP's & Topographies



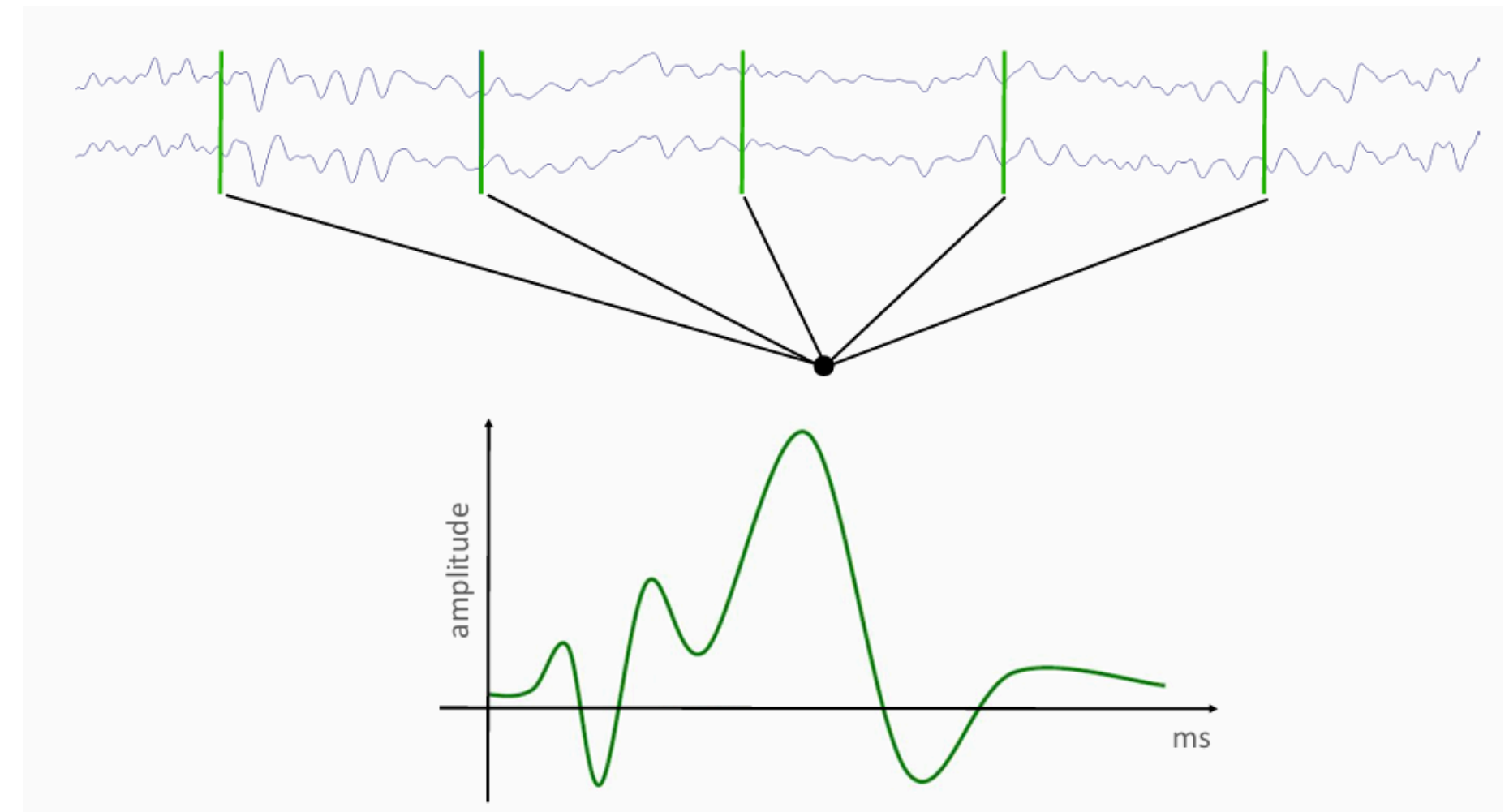
Time Domain: Event-related potentials

What is an ERP?

... an **electrical potential** related to a specific **event** (e.g. stimulus presentation)

- **mean** of EEG signal for a specified (event-related) time window => based on one/many channels, trials and participants

=> **data averaging over many Trials for every epoch!**



ERP's

What are we looking at in the Go/NoGo EEG?

Stimulus-locked ERPs

Time-locked to stimulus onset (Go or NoGo)

Response-locked ERPs

Time-locked to button press or inhibition

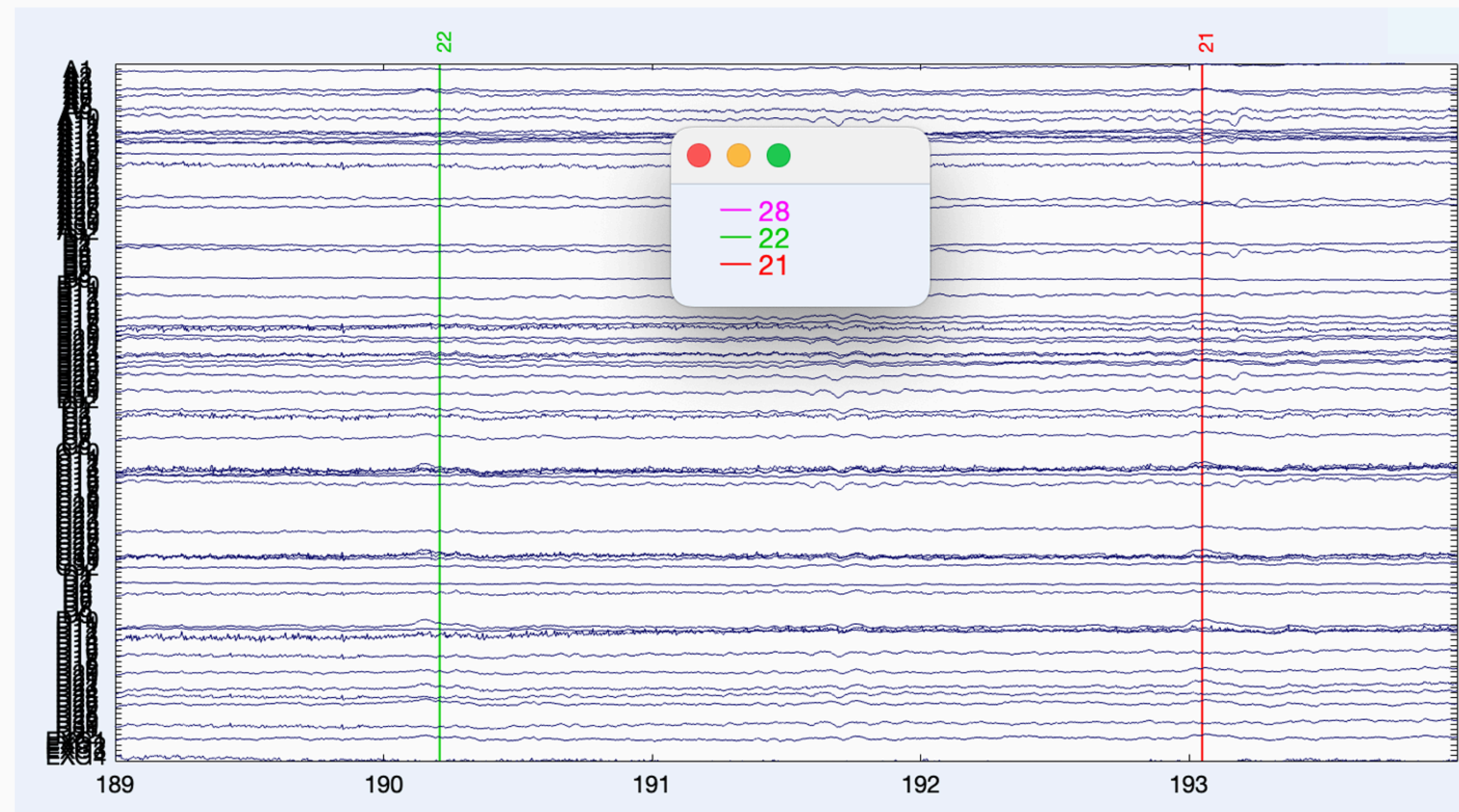
Key components of interest:

- N1/P1 → early visual processing
- N2 → conflict detection (frontocentral)
- P3 → response inhibition / evaluation
- ERN → error-related negativity

Analysis Pipeline

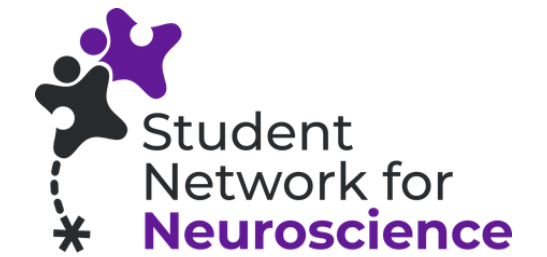
- 1 **Epoch the data** around stimulus onset
- 2 **Average epochs** per condition (Go / NoGo)
- 3 **Baseline correct** pre-stimulus window
- 4 **Identify peaks** amplitude & latency
- 5 **Compare conditions** Go vs. NoGo amplitudes

Trigger/Marker



trigger/marker =
time stamp for ERP
analysis

ERP naming system

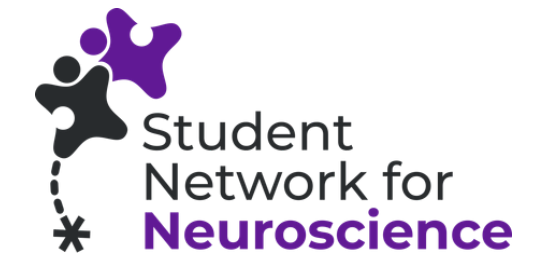


Naming system:

- N = negative and P = positive
- number = ms of peak latency after stimulus
or position of the peak in chronological order of the waveform
- sometimes functional names

Luck, S. J. (2014). Overview of Common ERP Components. In S. J. Luck (Ed.), *An Introduction to the Event-Related Potential Technique* (2nd ed., pp. 71 -118). Bradford Books.

Important ERPs



Exogenous - sensory components → triggered by the presence of a stimulus (partially modulated by top-down processes)

- **N1/N170** => primary sensory processing; N170 for visual face perception
- **Mismatch Negativity (MMN)** => detect deviations/mismatches from preceding to new (auditory) stimuli

Endogenous - reflect entirely task-dependent neural processes

- **P3 (P3a and P3b)** => novelty/unpredictable, surprising stimuli
- language-related: **N400** => mismatch of semantic information/violation of semantic expectancies

Response-related

- readiness potential => preparation for movement
- error-related negativity => negative-going deflection after error

Luck, S. J. (2014). Overview of Common ERP Components. In S. J. Luck (Ed.), *An Introduction to the Event-Related Potential Technique* (2nd ed., pp. 71 -118). Bradford Books.

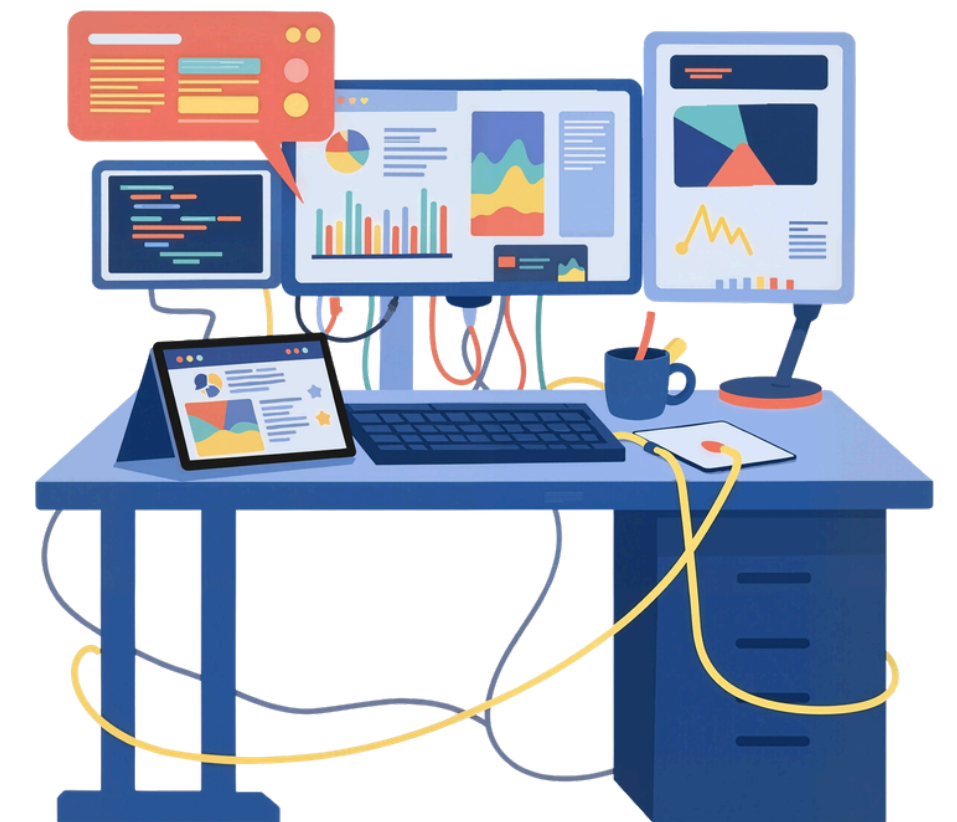
Go/No-go task and EEG

frontocentral ERPs in no-go trials:

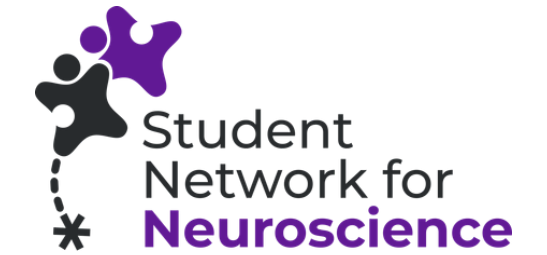
- no-go N200 → conflict detection / response monitoring
- no-go P300 → response inhibition / evaluation of action

Typical characteristics:

- strongest over frontocentral electrodes (e.g. Fz, Cz)
- larger amplitudes in No-Go vs Go trials



Component Analysis



Quantifying ERP Amplitudes and Latencies

What do we analyse?

- peak amplitude
- peak latency
- mean amplitude

chapter 9 in : Luck, S. J. (2014). An Introduction to the Event-Related Potential Technique (2nd ed.). Bradford Books

Component Analysis

Peak Amplitude & Latency

- Peak amplitude: the maximum (or minimum) voltage value of a component within a defined time window.
- Peak latency: the time point (ms) at which the peak occurs after stimulus onset.
- Advantage: easy to compute & interpret.
- Disadvantage: sensitive to noise — a single outlier trial can shift the peak.

Mean Amplitude

- Average voltage across all time points within a predefined window (e.g., 200–350 ms for N2).
- Advantage: more robust to noise and small waveform distortions.
- Disadvantage: requires an a priori window selection based on prior literature.
- Recommended for most ERP analyses (Luck, 2014, Ch. 9).

Luck, S. J. (2014). *An Introduction to the Event-Related Potential Technique* (2nd ed.), Ch. 9. Bradford Books.

In practice: inspect waveforms visually first, then select the method that best matches your research question.

ERPs - Go-/No-go task

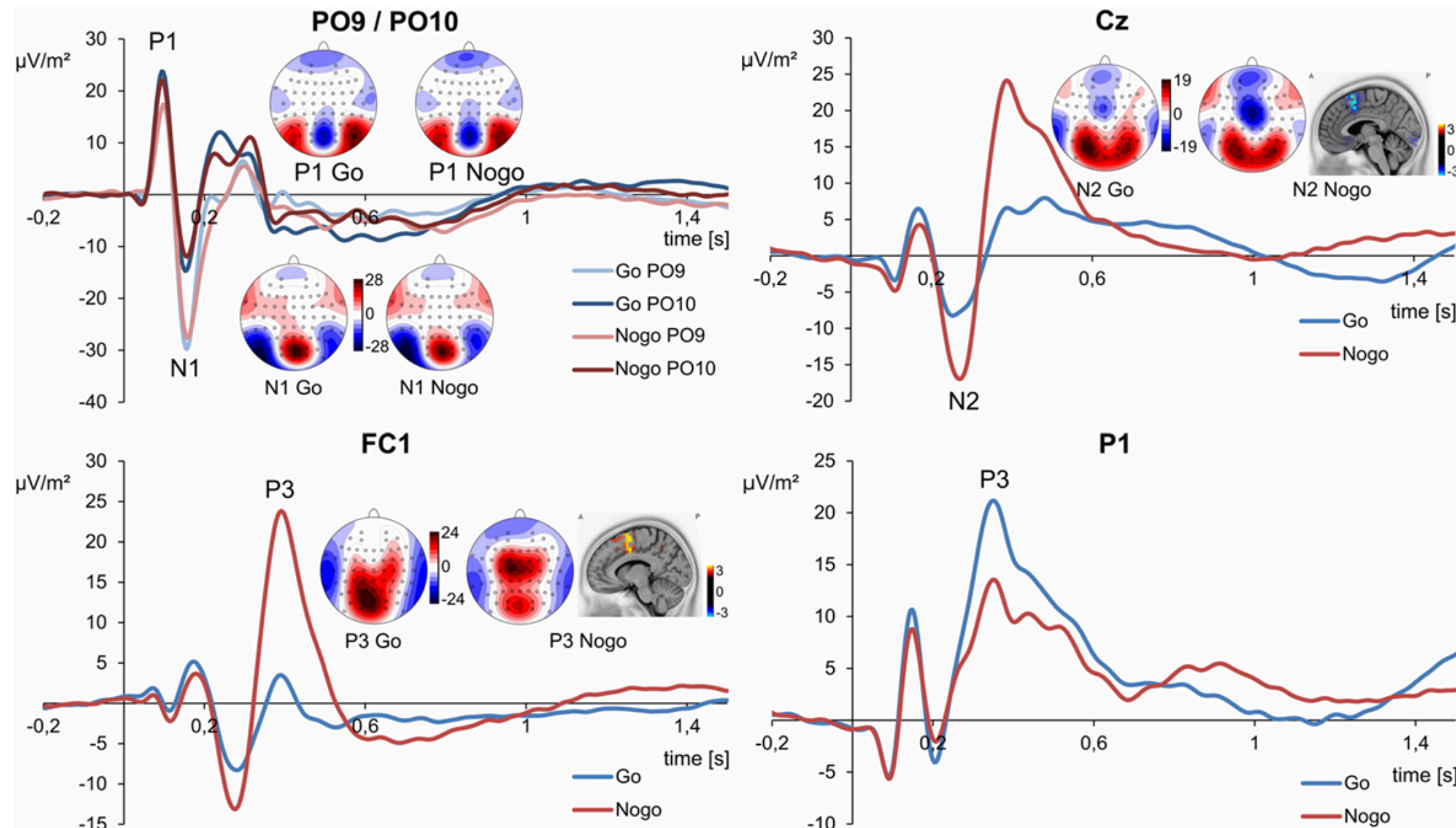


Figure 1. Event-related potential (ERP) components on Go and Nogo stimuli presentations. Plots are given for electrodes PO9/PO10 depicting P1 and N1 ERP, electrode Cz for N2, electrode FC1 for Nogo-P3 as well as electrode P1 for Go-P3. ERPs on Go stimuli are shown in blue, ERPs on Nogo stimuli are shown in red. The scalp topography plots reveal the distribution of voltages at the time point of the peak maximum of each ERP component. Time point zero denotes the time point of stimulus delivery. For N2 and P3 ERP, the results of sLORETA source estimation are shown.

Vahid, A., Mückschel, M., Neuhaus, A., Stock, A. K., & Beste, C. (2018). Machine learning provides novel neurophysiological features that predict performance to inhibit automated responses. *Scientific reports*, 8(1), 16235.

Topographies

What is a topographic map?

Spatial distribution of electrical activity

Voltage values interpolated across all electrodes at a given time point

How to read a topography:

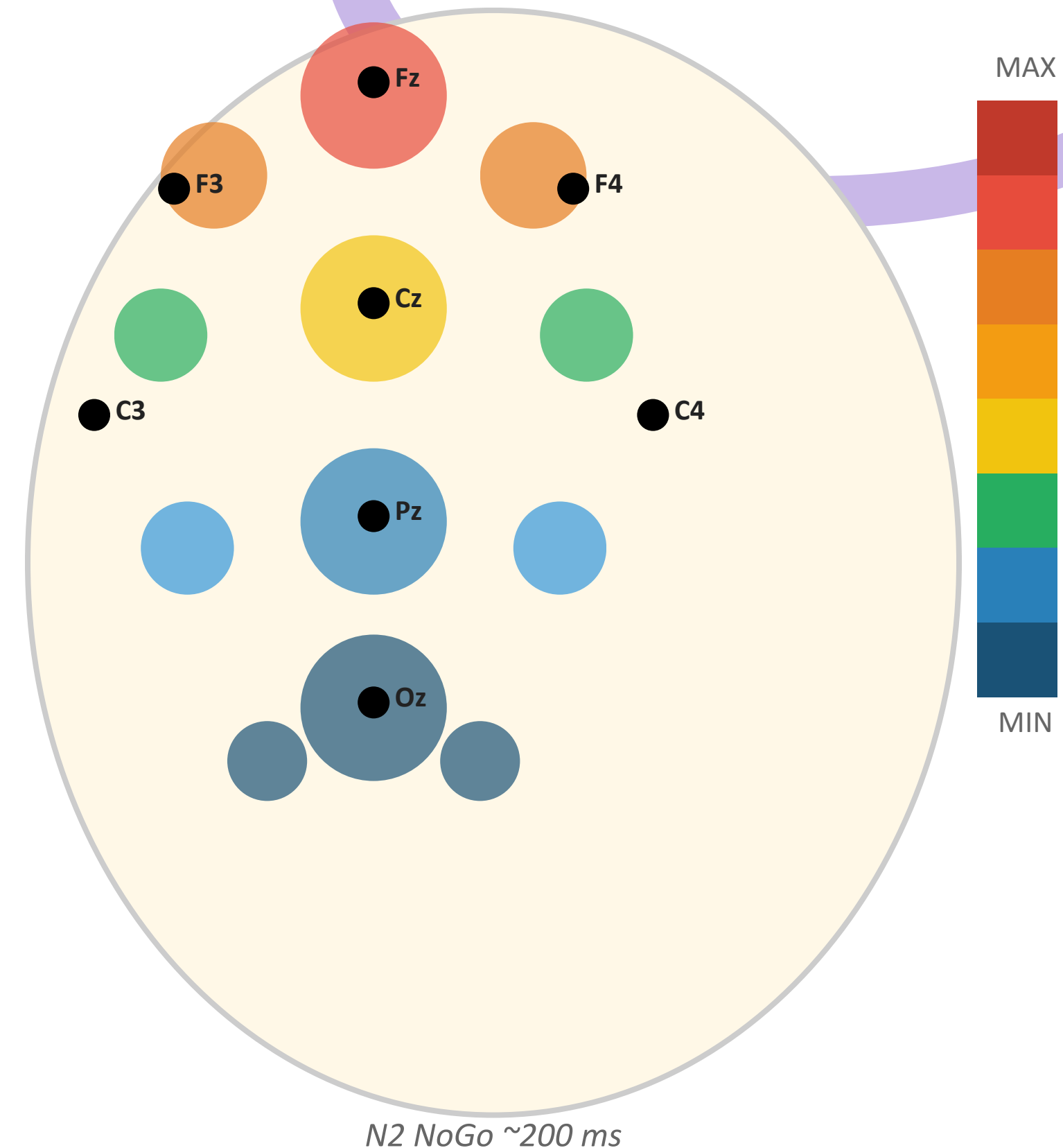
- Red = high positive voltage
- Blue = high negative voltage
- White/yellow = ~zero

When to plot topographies:

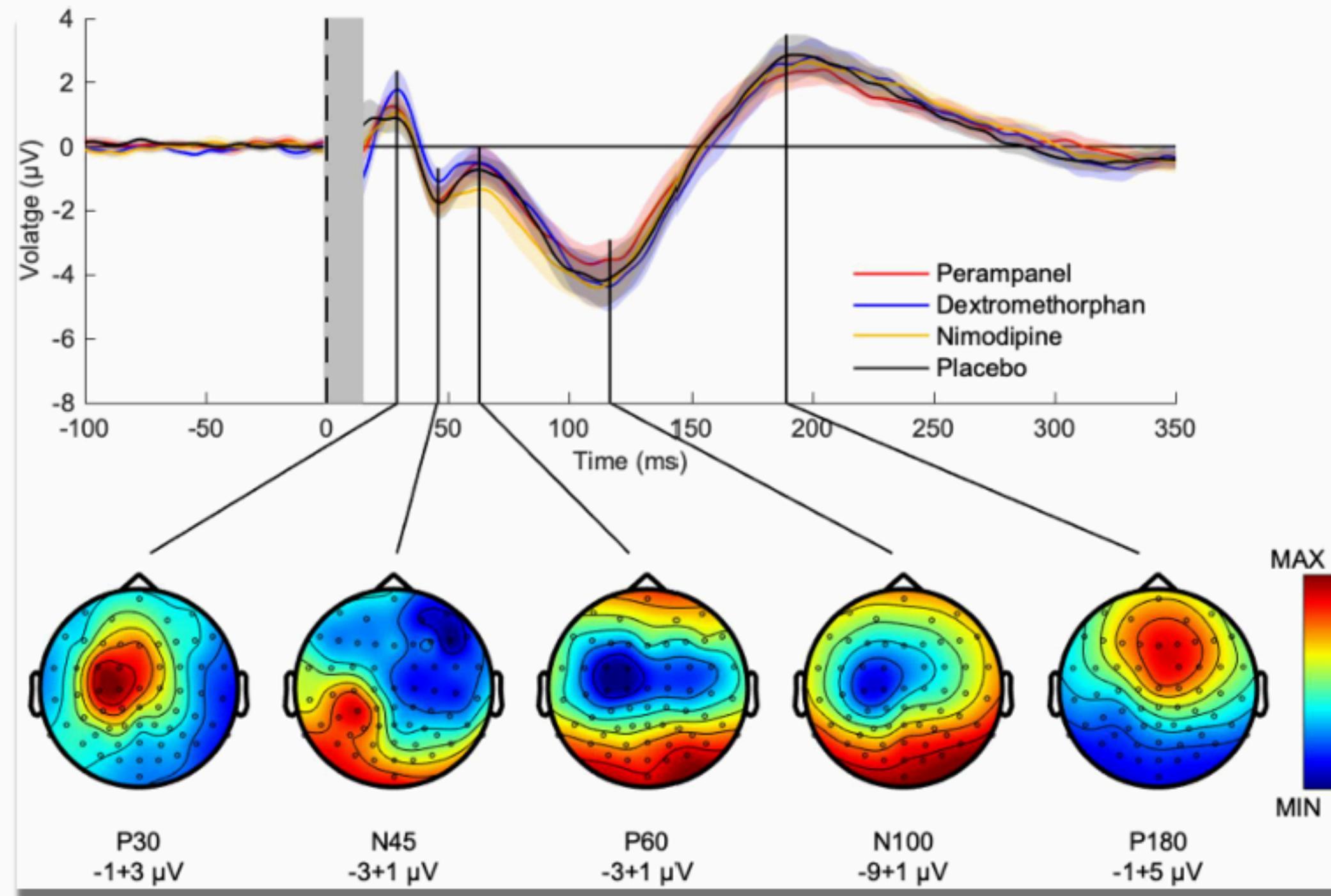
- At peak latency of a component (e.g., N2 peak ~200 ms)
- Mean amplitude within a time window
- Difference maps: NoGo minus Go

In MNE-Python:

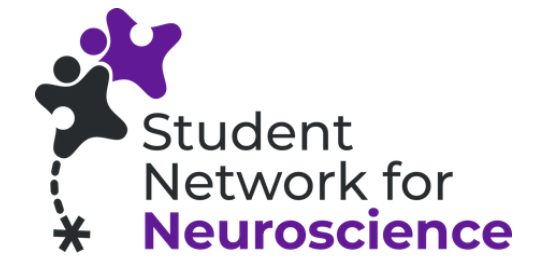
```
evoked.plot_topomap(times=[0.2, 0.3, 0.4])
```



What do the results look like?



Statistical Analysis - your task



```
# Define the Region of Interest (ROI)
channel_of_interest = ???
t_start = 0.250
t_end = 0.350

# Extract data for the ROI
###: Remove the extraction and averaging lines for the exercise
# pick() selects the channel, crop() selects the time window, get_data() converts to numpy array
go_data = epochs['Go'] ???
nogo_data = epochs['NoGo']. ???

# Average across the time window (the last dimension, axis=-1) to get one single voltage value per trial
go_means = np. ???
nogo_means = np. ???

# 3. Perform the Independent T-Test
t_stat, p_val = stats. ??? # Welch's t-test
```

Note: Pay attention to the scale! MNE uses volt → multiply by 1000000 to get μV

Statistical Analysis

ANOVA

Most common approach for ERP data.

Typical design

2×2 repeated-measures ANOVA

- Factor 1: Condition (Go vs. NoGo)
- Factor 2: Electrode (Fz, Cz, Pz)

Sphericity

Violations → apply Greenhouse–Geisser correction.

Correlation

Purpose

Relate ERP amplitudes to behavioural measures.

Examples

N2 amplitude ↔ commission error rate

P3 amplitude ↔ sensitivity d'

ERN amplitude ↔ RT after errors

Report

Use Pearson r or Spearman ρ .

Always report effect size r^2 alongside p .

Additional methods

t-test

Compare two conditions or groups
(e.g., NoGo N2: Go vs. NoGo).

Permutation / cluster tests

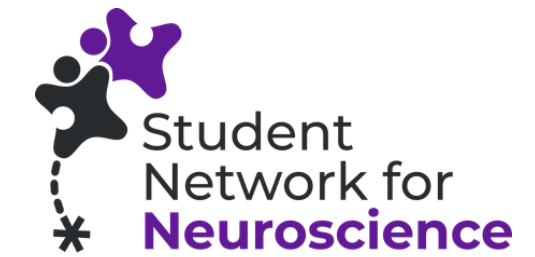
Mass-univariate approach;
corrects for multiple comparisons across time.

⚠ Correction

Always apply multiple comparison correction
(Bonferroni, FDR, or cluster-based).

=> chapter 10 in: Luck, S. J. (2014). An Introduction to the Event-Related Potential Technique (2nd ed.). Bradford Books

Interpretation – Relation Behavior and EEG



How do behavioral and neural measures connect?

Behavioral → EEG

Commission errors (CE) & ERN amplitude

Higher CE rate → larger ERN → poor inhibition

Reaction time & N2 latency

Slower RT → later N2 peak → more processing time needed

Sensitivity d' & P3 amplitude

Higher d' → larger NoGo-P3 → better inhibitory control

Response criterion β & frontal activation

Cautious responders show larger NoGo-N2

Statistical Approach

Correlation analysis

Pearson/Spearman r between ERP amplitude and behavioral measure (e.g., d' vs. P3)

ANOVA / t-test

Compare ERP amplitudes for correct vs. incorrect trials

Regression models

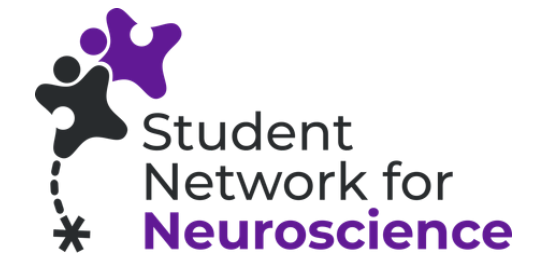
Predict behavioral outcome from ERP component features

Caution:

Correlation $\neq$ causation; consider multiple comparisons correction

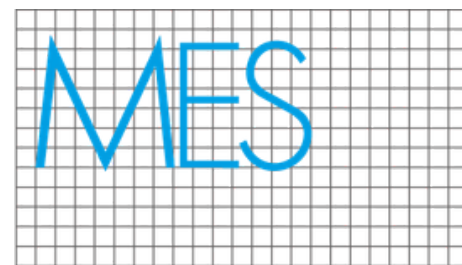
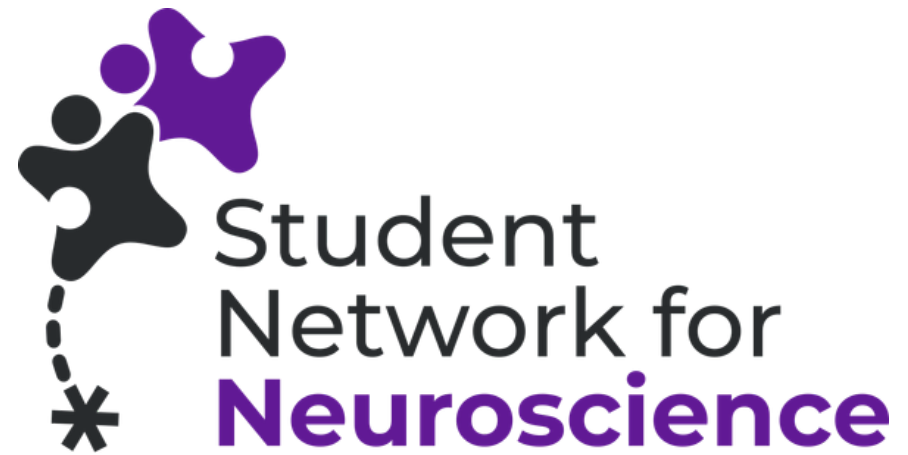
→ Reference: Luck, S. J. (2014), Ch. 10 – Statistical Analysis of ERP Data

An Outlook to more Complex Analysis methods



- For ERPs: Mass Univariate Analysis
- Time-Frequency Analysis
- Microstate Analysis
- Source Localization

chapter 9 in : Luck, S. J. (2014). An Introduction to the Event-Related Potential Technique (2nd ed.). Bradford Books



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Anwendungen



Any other questions?

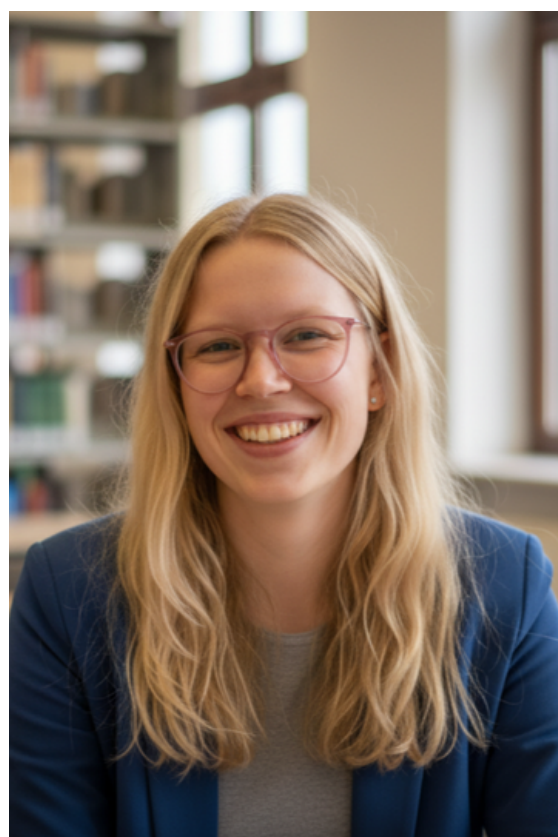


Thank you for your attention!



**More About
SNN**

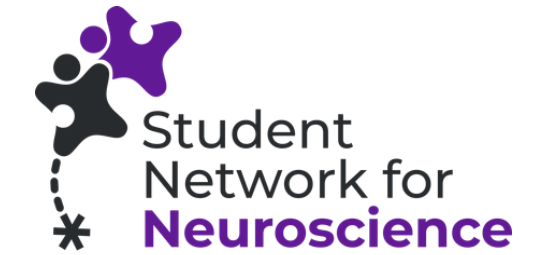
**Huge thanks
to the team
members
behind the
scenes!**



**More About
SNN**

Tutorials about MNE - including EVERYTHING!!
https://mne.tools/1.7/auto_tutorials/index.html

Literature and Sources



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- slides Merle Sagehorn (University of Osnabrück)
- <https://www.testable.org/experiment-guides/executive-function/go-no-go-task>

figures:

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- slides Merle Sagehorn (University of Osnabrück)

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- <https://dundeemedstudentnotes.wordpress.com/2012/04/12/temporal-and-spatial-summation/>
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